Experimental Probability

Name _____ Date _____ Calculator allowed Version 1

KEY WORDS experiment · trial · frequency · relative frequency · estimate · theoretical probability · fair

A Relative frequency ●○○ — ANSWER ON THE SHEET

Remember: Experimental probability = the number of times the event happened ÷ the total number of trials.

1. A spinner with four colours is spun many times. The table shows the results.

Colour	Red	Blue	Green	Yellow
Frequency	18	12	6	24

a) How many times was the spinner spun? _____

b) Write the experimental probability of red as a fraction in its simplest form. ___

c) Write the experimental probability of blue as a decimal. ___

d) Write the experimental probability of not getting green as a fraction. ___

e) Write the experimental probability of red or yellow as a fraction. ___

B Estimating from experiments ●●○ — WORK IN YOUR BOOK

Remember: Use the results as an estimate of the probability. Then expected number = probability × number of trials.

2. A drawing pin is dropped 200 times. It lands point up 74 times.

a) Estimate the probability that it lands point up.

b) How many times would you expect it to land point up in 1000 drops?

3. Of the last 150 cars to pass a school, 45 were white.

a) Estimate the probability that the next car is white.

b) How many of the next 400 cars would you expect to be white?

4. A basketball player scored from 54 of her last 90 free throws.

a) Estimate the probability that she scores from her next free throw.

b) How many of her next 30 free throws would you expect her to score from?

5. A factory tests 500 light bulbs and finds that 15 are faulty.

a) Estimate the probability that a bulb is faulty.

b) How many faulty bulbs would you expect in a batch of 2000?

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C Experiment and theory ●●● — WORK IN YOUR BOOK

⚠ **Watch out:** Results from an experiment are never exactly what theory predicts. Look for results that are a long way from what is expected.

6. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	9	11	10	8	12	10

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

7. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	6	7	8	7	8	24

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

D More trials ●●● — ANSWER ON THE SHEET

💡 **Remember:** The more trials there are, the closer the experimental probability usually gets to the true probability.

8. A coin is tossed again and again, and the number of heads so far is recorded. Complete the table. Relative frequency = number of heads ÷ number of tosses.

Number of tosses	10	50	100	500	1000
Number of heads	7	29	54	256	497
Relative frequency					

9. Use your table.

a) Which relative frequency is the best estimate of the probability of a head? Give a reason.

b) Does the table suggest that the coin is fair? Explain.

★ Extension — FOR EARLY FINISHERS

10. A scientist catches 40 fish from a lake, tags them and lets them go. Later she catches 50 fish and finds that 8 of them have tags. Estimate the number of fish in the lake.

11. A bag holds 20 counters. A counter is picked and put back 100 times, and it is red 35 times. Estimate the number of red counters in the bag.

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Experimental Probability

Name _____ Date _____ Calculator allowed Version 2

KEY WORDS experiment · trial · frequency · relative frequency · estimate · theoretical probability · fair

A Relative frequency ●○○ — ANSWER ON THE SHEET

Remember: Experimental probability = the number of times the event happened ÷ the total number of trials.

1. A spinner with four colours is spun many times. The table shows the results.

Colour	Red	Blue	Green	Yellow
Frequency	14	11	5	20

a) How many times was the spinner spun? _____

b) Write the experimental probability of red as a fraction in its simplest form. ___

c) Write the experimental probability of blue as a decimal. ___

d) Write the experimental probability of not getting green as a fraction. ___

e) Write the experimental probability of red or yellow as a fraction. ___

B Estimating from experiments ●●○ — WORK IN YOUR BOOK

Remember: Use the results as an estimate of the probability. Then expected number = probability × number of trials.

2. A drawing pin is dropped 200 times. It lands point up 86 times.

a) Estimate the probability that it lands point up.

b) How many times would you expect it to land point up in 1000 drops?

3. Of the last 120 cars to pass a school, 30 were white.

a) Estimate the probability that the next car is white.

b) How many of the next 400 cars would you expect to be white?

4. A basketball player scored from 56 of her last 80 free throws.

a) Estimate the probability that she scores from her next free throw.

b) How many of her next 30 free throws would you expect her to score from?

5. A factory tests 500 light bulbs and finds that 20 are faulty.

a) Estimate the probability that a bulb is faulty.

b) How many faulty bulbs would you expect in a batch of 2000?

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C Experiment and theory ●●● — WORK IN YOUR BOOK

⚠ **Watch out:** Results from an experiment are never exactly what theory predicts. Look for results that are a long way from what is expected.

6. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	11	9	12	10	8	10

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

7. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	7	22	8	6	9	8

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

D More trials ●●● — ANSWER ON THE SHEET

💡 **Remember:** The more trials there are, the closer the experimental probability usually gets to the true probability.

8. A coin is tossed again and again, and the number of heads so far is recorded. Complete the table. Relative frequency = number of heads ÷ number of tosses.

Number of tosses	10	50	100	500	1000
Number of heads	3	22	47	244	503
Relative frequency					

9. Use your table.

a) Which relative frequency is the best estimate of the probability of a head? Give a reason.

b) Does the table suggest that the coin is fair? Explain.

★ Extension — FOR EARLY FINISHERS

10. A scientist catches 60 fish from a lake, tags them and lets them go. Later she catches 40 fish and finds that 6 of them have tags. Estimate the number of fish in the lake.

11. A bag holds 40 counters. A counter is picked and put back 200 times, and it is red 45 times. Estimate the number of red counters in the bag.

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